Q1 (4) - Single Correct

$0=(\vec{a}+\vec{b})\cdot(2\vec{a}+3\vec{b})\times(3\vec{a}-2\vec{b})=(\vec{a}+\vec{b})\cdot(-4\vec{a}\times\vec{b}-9\vec{a}\times\vec{b})=-13(\vec{a}+\vec{b}).(\vec{a}\times\vec{b})$

which is true for all values of $\vec{a}$ and $\vec{b}$.

Q2 (4) - Single Correct

Volume of the parallelopiped formed by $\vec{a}', \vec{b}', \vec{c}'$ is 4

$\therefore$ Volume of the parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ is $\frac{1}{4}$

$\overrightarrow{\mathrm{b}}\times\overrightarrow{\mathrm{c}}=\frac{\left(\overrightarrow{\mathrm{c}}'\times\overrightarrow{\mathrm{a}}'\right)\times\overrightarrow{\mathrm{C}}}{4}=\frac{1}{4}\overrightarrow{\mathrm{a}}'$

$|\vec{b}\times\vec{c}|=\frac{\sqrt{2}}{4}=\frac{1}{2\sqrt{2}}$

$\therefore$ length of altitude $=\frac{1}{4}\times 2\sqrt{2}=\frac{1}{\sqrt{2}}$.

Q3 (2) - Single Correct

Let $\overrightarrow{\mathrm{a}}=\lambda\overrightarrow{\mathrm{b}}+\mu\overrightarrow{\mathrm{c}}$

then $\frac{\overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{b}}}{\mathrm{ab}}=\frac{\overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{d}}}{\mathrm{ad}}$

i.e. $\frac{(\lambda\overrightarrow{\mathrm{b}}+\mu\overrightarrow{\mathrm{c}})\cdot\overrightarrow{\mathrm{b}}}{\mathrm{b}}=\frac{(\lambda\overrightarrow{\mathrm{b}}+\mu\overrightarrow{\mathrm{c}})\cdot\overrightarrow{\mathrm{d}}}{\mathrm{d}}$

i.e. $\frac{[\lambda(2\hat{\mathrm{i}}+\hat{\mathrm{j}})+\mu(\hat{\mathrm{i}}-\hat{\mathrm{j}}+\hat{\mathrm{k}})]\cdot(2\hat{\mathrm{i}}+\hat{\mathrm{j}})}{\sqrt{5}}=\frac{[\lambda(2\hat{\mathrm{i}}+\hat{\mathrm{j}})+\mu(\hat{\mathrm{i}}-\hat{\mathrm{j}}+\hat{\mathrm{k}})]\cdot(\hat{\mathrm{j}}+2\hat{\mathrm{k}})}{\sqrt{5}}$

$\therefore$ $\hat{\mathrm{a}}=\frac{\hat{\mathrm{i}}-\hat{\mathrm{j}}+\hat{\mathrm{k}}}{\sqrt{3}}$

Q4 (3) - Single Correct

Since $\vec{a}_1, \vec{a}_2 \& \vec{a}_3$ are non-coplanar vectors

$\therefore x+y-3=0$ ...(i)

$2x-y+2=0$ ...(ii)

$2x+y+\lambda=0$ ...(iii)

From (i) & (ii) $x=1/3, y=8/3$

$\therefore$ from (iii) $\lambda=-\frac{10}{3}$

Q5 (2) - Single Correct

$\vec{c} = \lambda(\vec{a} \times \vec{b})$

$1 = (\vec{a} \times \vec{b}) \cdot \vec{c} = \frac{\vec{c} \cdot \vec{c}}{\lambda} = \frac{|\vec{c}|^2}{\lambda} = \frac{1}{3\lambda}$

$\lambda = \frac{1}{3}$

$\vec{c}^2 = \lambda^2(\vec{a} \times \vec{b})^2$

$\frac{1}{3} = \frac{1}{9}\left(a^2 b^2 \sin^2\theta\right) = \frac{1}{9} \times 2 \times 3 \sin^2\theta$

$\sin^2\theta = \frac{1}{2} \quad \therefore \quad \theta = \frac{\pi}{4}$

Q6 (3) - Single Correct

Plane $4x + 3y + 2 = 0 \quad \ldots\ldots$ (i)

$z = 0 \ldots\ldots\ldots\ldots$ (ii)

Any plane passing through the intersecting line of the given planes is $4x + 3y + \lambda z + 2 = 0 \ldots\ldots\ldots\ldots$ (iii)

This for suitable value of λ will be inclined at 45° to the plane (i) Direction cosine of the normal to the plane (iii) are -

$\left(\frac{4}{\sqrt{25+\lambda^2}}, \frac{3}{\sqrt{25+\lambda^2}}, \frac{\lambda}{\sqrt{25+\lambda^2}}\right)$

Direction cosines of the normal to the plane (i) are

$\left(\frac{4}{5}, \frac{3}{5}, \frac{0}{5}\right)$

$\cos 45^\circ = \frac{16+9+0}{5\sqrt{25+\lambda^2}}$

$\Rightarrow \lambda = \pm 5$

$\therefore$ the required plane is $4x + 3y + 5z - 2 = 0$ Length of perpendicular from the origin to the above plane $= \frac{2}{\sqrt{50}} = \frac{\sqrt{2}}{5}$

Q7 (2) - Single Correct

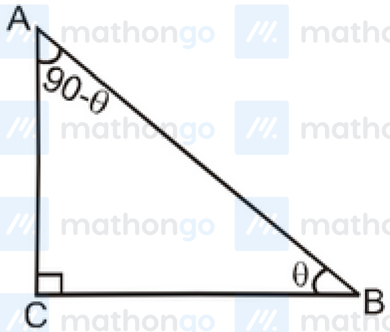
We have, $\overrightarrow{AB} \cdot \overrightarrow{AC} = |\overrightarrow{AB}||\overrightarrow{AC}| \cos(90^\circ - \theta) = AB \cdot AC \sin\theta$

$\overrightarrow{BC} \cdot \overrightarrow{BA} = |\overrightarrow{BC}| \cdot |\overrightarrow{BA}| \cos\theta = BC \cdot BA \cos\theta$

$\overrightarrow{CA} \cdot \overrightarrow{CB} = |\overrightarrow{CA}| \cdot |\overrightarrow{CB}| \cos 90^\circ = 0$

$\therefore \quad \overrightarrow{AB} \cdot \overrightarrow{AC} + \overrightarrow{BC} \cdot \overrightarrow{BA} + \overrightarrow{CA} \cdot \overrightarrow{CB} = AB \cdot AC \sin\theta + BC \cdot BA \cos\theta$

$= \mathrm{AB}\left[\mathrm{AC} \cdot \frac{\mathrm{AC}}{\mathrm{AB}} + \mathrm{BC} \cdot \frac{\mathrm{BC}}{\mathrm{AB}}\right] = \mathrm{AC}^2 + \mathrm{BC}^2 = \mathrm{AB}^2 = \mathrm{p}^2$



Q8 (1) - Single Correct

Foot of perpendicular from point $\mathrm{A}(\vec{a})$ on the plane $\vec{r} \cdot \vec{n} = \mathrm{d}$ is $\vec{a} + \frac{(\mathrm{d} - \vec{a} \cdot \vec{n})}{|\vec{n}|^2}\vec{n}$

$\therefore$ Equation of line parallel to $\vec{r} = \vec{a} + \lambda\vec{b}$ in the plane $\vec{r} \cdot \vec{n} = d$ is given by

$\vec{r} = \vec{a} + \frac{(d - \vec{a} \cdot \vec{n})}{|\vec{n}|^2}\vec{n} + \lambda\vec{b}$

Q9 (2) - Single Correct

Let direction ratios of the line be $\langle a, b, c\rangle$, then

$2a - b + c = 0$

$a - b - 2c = 0$

i.e. $\frac{a}{3} = \frac{b}{5} = \frac{c}{-1}$

$\therefore$ direction ratios of the line are $\langle 3, 5, -1\rangle$

Any point on the line is $(2+\lambda, 2-\lambda, 3-2\lambda)$. It lies on the plane π if $2(2+\lambda) - (2-\lambda) + (3-2\lambda) = 4$

i.e. $4 + 2\lambda - 2 + \lambda + 3 - 2\lambda = 4$

i.e. $\lambda = -1$

$\therefore$ The point of intersection of the line and the plane is $(1, 3, 5)$

$\therefore$ equation of the required line is $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

Q10 (2) - Single Correct

Let a point $(3\lambda+1, \lambda+2, 2\lambda+3)$ of the first line also lies on the second line.

Then $\frac{3\lambda+1-3}{1} = \frac{\lambda+2-1}{2} = \frac{2\lambda+3-2}{3} \Rightarrow \lambda = 1$

Hence the point of intersection P of the two lines is $(4,3,5)$ Equation of plane perpendicular to OP where O is $(0,0,0)$ and passing through P is

$4x+3y+5z=50$

Q11 (2) - Single Correct

$1=\left|(\overrightarrow{\mathrm{b}}-\overrightarrow{\mathrm{a}})\cdot\frac{(\overrightarrow{\mathrm{p}}\times\overrightarrow{\mathrm{q}})}{|\overrightarrow{\mathrm{p}}\times\overrightarrow{\mathrm{q}}|}\right| \Rightarrow |\overrightarrow{\mathrm{b}}-\overrightarrow{\mathrm{a}}|\cos 60^{\circ}=\frac{1}{2}\mathrm{AB} \quad\Rightarrow\quad \mathrm{AB}=2$

Q12 (4) - Single Correct

$P_1=P_2=0, P_2=P_3=0$ and $P_3=P_1=0$ are lines of intersection of the three planes P_1, P_2 and P_3. As $\vec{n}_1, \vec{n}_2$ and $\vec{n}_3$ are non-coplanar, planes P_1, P_2 and P_3 will intersect at unique point. So the given lines will pass through a fixed point.

Q13 (1) - Single Correct

Let θ be the required angle then θ will be the angle between $\vec{a}$ and $\vec{b}+\vec{c}(\vec{b}+\vec{c}$ lies along the angular bisector of $\vec{a}$ and $\vec{b})$

$$\cos\theta=\frac{\vec{a}\cdot(\vec{b}+\vec{c})}{|\vec{a}||\vec{b}+\vec{c}|}$$

$$=\frac{2\cos\alpha}{\sqrt{2+2\cos\alpha}}=\frac{\cos\alpha}{\cos\frac{\alpha}{2}}$$

$$\theta=\cos^{-1}\left(\frac{\cos\alpha}{\cos\alpha/2}\right)$$

Q14 (1) - Single Correct

$A(1,1,1), B(2,3,5), C(-1,0,2)$ directions ratios of AB are $<1,2,4>$ direction ratios of AC are $<-2,-1,1>\therefore$ direction ratios of normal to plane ABC are $<2,-3,1>$

$\therefore$ Equation of the plane ABC is $2x-3y+z=0$

Let the equation of the required plane be $2x-3y+z=k$, then $\left|\frac{k}{\sqrt{4+9+1}}\right|=2$

$\mathrm{k}=\pm2\sqrt{14}$

$\therefore\quad$ Equation of the required plane is $2x-3y+z+2\sqrt{14}=0$

Q15 (1,3) - Multiple Correct

$(\vec{a}-\vec{b}) \times[(\vec{b}+\vec{a}) \times(2 \vec{a}+\vec{b})]=\vec{b}+\vec{a}$

$\Rightarrow \quad\{(\vec{a}-\vec{b}) \cdot(2 \vec{a}+\vec{b})\}(\vec{b}+\vec{a})-\{(\vec{a}-\vec{b}) \cdot(\vec{b}+\vec{a})\}(2 \vec{a}+\vec{b})=\vec{b}+\vec{a}$

$\Rightarrow \quad(2-\vec{a} \cdot \vec{b}-1)(\vec{b}+\vec{a})=\vec{b}+\vec{a}$

$\Rightarrow \quad$ either $\overrightarrow{\mathrm{b}}+\overrightarrow{\mathrm{a}}=\overrightarrow{0}$ or $1-\overrightarrow{\mathrm{a}} \cdot \overrightarrow{\mathrm{b}}=1$

$\Rightarrow \quad$ either $\overrightarrow{\mathrm{b}}=-\overrightarrow{\mathrm{a}}$ or $\overrightarrow{\mathrm{a}} \cdot \overrightarrow{\mathrm{b}}=0$

$\Rightarrow \quad$ either $\theta=\pi$ or $\theta=\frac{\pi}{2}$

Q16 (1,2,4) - Multiple Correct

$(\lambda-1)\left(\vec{a}_1-\vec{a}_2\right)+\mu\left(\vec{a}_2+\vec{a}_3\right)+\gamma\left(\vec{a}_3+\vec{a}_4-2 \vec{a}_2\right)+\vec{a}_3+\delta \vec{a}_4=\overrightarrow{0}$

i.e. $\quad(\lambda-1) \overrightarrow{\mathrm{a}}_1+(1-\lambda+\mu-2 \gamma) \overrightarrow{\mathrm{a}}_2+(\mu+\gamma+1) \overrightarrow{\mathrm{a}}_3+(\gamma+\delta) \overrightarrow{\mathrm{a}}_4=\overrightarrow{0}$

Since $\overrightarrow{\mathrm{a}}_1, \overrightarrow{\mathrm{a}}_2, \overrightarrow{\mathrm{a}}_3, \overrightarrow{\mathrm{a}}_4$ are linearly independent

$\therefore \lambda-1=0,1-\lambda+\mu-2 \gamma=0, \mu+\gamma+1=0$ and $\gamma+\delta=0$

i.e. $\quad \lambda=1, \mu=2 \gamma, \mu+\gamma+1=0, \gamma+\delta=0$

i.e. $\quad \lambda=1, \mu=-\frac{2}{3}, \gamma=-\frac{1}{3}, \delta=\frac{1}{3}$

Q17 (3,4) - Multiple Correct

Since $[\vec{a} \vec{b} \vec{c}]=0$

$\therefore \quad \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar vectors.

Further since $\vec{d}$ is equally inclined to $\vec{a}, \vec{b}$ and $\vec{c}$

$\therefore \quad \overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{a}}=\overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{b}}=\overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{c}}=0$

$\therefore \quad \overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{x}}=\overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{y}}=\overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{z}}=0$

$\therefore \quad \overrightarrow{\mathrm{d}} \cdot \overrightarrow{\mathrm{r}}=0$

Q18 (1,3,4) - Multiple Correct

(1) $\quad \overrightarrow{\mathrm{a}} \times[\overrightarrow{\mathrm{a}} \times(\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}})]=\overrightarrow{\mathrm{a}} \times[(\overrightarrow{\mathrm{a}} \cdot \overrightarrow{\mathrm{b}}) \overrightarrow{\mathrm{a}}-(\overrightarrow{\mathrm{a}} \cdot \overrightarrow{\mathrm{a}}) \overrightarrow{\mathrm{b}}]$

$=-(\vec{a} \cdot \vec{a})(\vec{a} \times \vec{b})$

$\therefore \quad$ (1) is not correct

(2) $\overrightarrow{\mathrm{v}} \cdot \overrightarrow{\mathrm{a}} = \overrightarrow{0} \quad \Rightarrow \quad \overrightarrow{\mathrm{v}} = \overrightarrow{0}$ or $\overrightarrow{\mathrm{v}} \perp \overrightarrow{\mathrm{a}}$

$\overrightarrow{\mathrm{v}} \cdot \overrightarrow{\mathrm{b}} = \overrightarrow{0} \quad \Rightarrow \quad \overrightarrow{\mathrm{v}} = \overrightarrow{0}$ or $\overrightarrow{\mathrm{v}} \perp \overrightarrow{\mathrm{b}}$

$\overrightarrow{\mathrm{v}} \cdot \overrightarrow{\mathrm{c}} = \overrightarrow{0} \quad \Rightarrow \quad \overrightarrow{\mathrm{v}} = \overrightarrow{0}$ or $\overrightarrow{\mathrm{v}} \perp \overrightarrow{\mathrm{c}}$

$\therefore \quad \overrightarrow{\mathrm{v}} = \overrightarrow{0}$ or $\overrightarrow{\mathrm{v}} \perp \overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{b}}, \overrightarrow{\mathrm{c}}$

$\therefore \quad \overrightarrow{\mathrm{v}} = \overrightarrow{0}$

(3) $(\overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}}) \cdot (\overrightarrow{\mathrm{c}} \times \overrightarrow{\mathrm{d}}) = \overrightarrow{0}$

$\therefore$ statement is incorrect

(4) $\vec{a} \times \vec{b}' + \vec{b} \cdot \vec{c}' + \vec{c} \cdot \vec{a}' = 0$. (Property of reciprocal system)

(4) is incorrect

Q19 (3,4) - Multiple Correct

Let $\vec{a}, \vec{b}, \vec{c}$ lie in the x -y plane

Let $\vec{a} = \hat{\imath}, \vec{b} = -\frac{1}{2}\hat{\imath} + \frac{\sqrt{3}}{2}\hat{\jmath}$ and $\vec{c} = -\frac{1}{2}\hat{\imath} - \frac{\sqrt{3}}{2}\hat{\jmath}$

$\therefore \quad |\vec{p} + \vec{q} + \vec{r}| = |\lambda\vec{a} + \mu\vec{b} + v\vec{c}| = \left|\lambda\hat{\imath} + \mu\left(-\frac{1}{2}\hat{\imath} + \frac{\sqrt{3}}{2}\hat{\jmath}\right) + v\left(-\frac{1}{2}\hat{\imath} - \frac{\sqrt{3}}{2}\hat{\jmath}\right)\right|$

$= \left|\left(\lambda - \frac{\mu}{2} - \frac{v}{2}\right)\hat{\imath} + \frac{\sqrt{3}}{2}(\mu - v)\hat{\jmath}\right| = \sqrt{\left(\lambda - \frac{\mu}{2} - \frac{v}{2}\right)^2 + \frac{3}{4}(\mu - v)^2} = \sqrt{\lambda^2 + \mu^2 + v^2 - \lambda\mu - \lambda v - \mu v}$

$= \frac{1}{\sqrt{2}}\sqrt{(\lambda - \mu)^2 + (\mu - v)^2 + (v - \lambda)^2} \geq \frac{1}{\sqrt{2}}\sqrt{1 + 1 + 4}$

$\geq \sqrt{3}$

$\therefore |\vec{p} + \vec{q} + \vec{r}|$ can take a value equal to $\sqrt{3}, 2$

Q20 (1,2,3) - Multiple Correct

$x + y + z - 1 = 0$

$4x + y - 2z + 2 = 0$

$\therefore$ direction ratios of the line are $< -3, 6, -3 >$

i.e. $< 1, -2, 1\rangle$

Let $\quad z = k$, then $x = k - 1, y = 2 - 2k$

i.e. $\quad (\mathrm{k} - 1, 2 - 2\mathrm{k}, \mathrm{k})$ is any point on the line

$\therefore (-1,2,0), (0,0,1)$ and $\left(-\frac{1}{2}, 1, \frac{1}{2}\right)$ are points on the line

$\therefore$ (1), (2) and (3) are correct options

Q21 (1,2) - Multiple Correct

$3x - 6y + 2z + 5 = 0 \quad \ldots\ldots \text{(i)}$

$-4x + 12y - 3z + 3 = 0 \quad \ldots\ldots\ldots \text{(ii)}$

$\frac{3x-6y+2z+5}{\sqrt{9+36+4}} = \frac{-4x+12y-3z+3}{\sqrt{16+144+9}}$

bisects the angle between the planes that contains the origin.

$13(3x - 6y + 2z + 5) = 7(-4x + 12y - 3z + 3)$

$39x - 78y + 26z + 65 = -28x + 84y - 21z + 21$

$67x - 162y + 47z + 44 = 0 \quad \ldots\ldots \text{(iii)}$

Further $3 \times (-4) + (-6)(12) + 2 \times (-3) < 0$

$\therefore$ origin lies in acute angle

Q22 (1,2) - Multiple Correct

Equation of required plane is $\ell x + my + \lambda z = 0 \quad \cdots \text{(i)}$

angle between (i) & $\ell x + my = 0$ is α.

$\Rightarrow \quad \cos\alpha = \frac{\ell^2+m^2}{\sqrt{\ell^2+m^2}\sqrt{\ell^2+m^2+\lambda^2}}$

$\Rightarrow \quad \cos^2\alpha = \frac{\ell^2+\mathrm{m}^2}{\ell^2+\mathrm{m}^2+\lambda^2} \Rightarrow \quad \lambda = \pm\sqrt{\ell^2+\mathrm{m}^2}\tan\alpha$

Hence equation of plane is

$\ell \mathrm{x} + \mathrm{my} \pm \mathrm{z}\sqrt{\ell^2+\mathrm{m}^2}\tan\alpha = 0$

Q23 (4) - Paragraph 1

shortest distance between both lines

$$= \left| \frac{\begin{matrix} 3-2 & 1-1 & 0+1 \\ 1 & 0 & 2 \\ 1 & 1 & -1 \end{matrix} \Bigg|}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2 \\ 1 & 1 & -1 \end{vmatrix}} \right| = \left| \frac{1(0-2)-0(-1-2)+1(1-0)}{-2\hat{i}+3\hat{j}+\hat{k}} \right| = \frac{1}{\sqrt{4+9+1}} = \frac{1}{\sqrt{14}}$$

Q24 (2) - Paragraph 1

Equation of plane P

$$\begin{vmatrix} x-2 & y-1 & z+1 \\ 1 & 0 & 2 \\ 1 & 1 & -1 \end{vmatrix} = 0$$

$-2(x-2)+(y-1)(3)+(z+1)=0$

$-2x+3y+z+2=0$

$2x-3y-z-2=0$

Now image of point $0(0,0,0)$ in plane P

$\frac{x-0}{2}=\frac{y-0}{-3}=\frac{z-0}{-1}=\frac{-2(-2)}{4+9+1}$

$\frac{x}{2}=\frac{y}{-3}=\frac{z}{-1}=\frac{4}{14}$

Image point $\left(\frac{4}{7}, \frac{-6}{7}, \frac{-2}{7}\right)$

Q25 (4) - Paragraph 2

$\mathrm{u}\ell+\mathrm{vm}+\mathrm{wn}=0$

$\mathrm{a}\ell^2+\mathrm{bm}^2+\mathrm{cn}^2=0$

$\mathrm{a}\ell^2+\mathrm{bm}^2+\mathrm{c}\left\{-\frac{(4\ell+\mathrm{vm})}{\mathrm{w}}\right\}^2=0$

$\Rightarrow\left(aw^2+cu^2\right)\ell^2+\left(bw^2+cv^2\right)m^2+2\text{ cuv }\ell m=0$

$\Rightarrow\left(\mathrm{aw}^2+\mathrm{cu}^2\right)\left(\frac{\ell}{\mathrm{m}}\right)^2+\left(\mathrm{bw}^2+\mathrm{cv}^2\right)+2\,\mathrm{cuv}\left(\frac{\ell}{\mathrm{m}}\right)=0 \ldots\ldots\ldots\ldots \text{(i)}$

put $u=v=w=1$ in equation, then

$(a+c)\left(\frac{\ell}{m}\right)^2+2c\left(\frac{\ell}{m}\right)+(b+c)=0$

similarly $(a+b)\left(\frac{m}{n}\right)^2+2a\left(\frac{m}{n}\right)+(c+a)=0$

and $(b+c)\left(\frac{n}{\ell}\right)^2+2b\left(\frac{n}{\ell}\right)+(a+b)=0 \ldots\ldots\ldots\ldots \text{(ii)}$

From equation

(ii) $\frac{n_1}{\ell_1}\cdot\frac{n_2}{\ell_2}=\frac{a+b}{b+c}$

similarly $\frac{\ell_1\ell_2}{\mathrm{b+c}}=\frac{\mathrm{m}_1\,\mathrm{m}_2}{\mathrm{c+a}}=\frac{\mathrm{n}_1\mathrm{n}_2}{\mathrm{a+b}} \ldots\ldots\ldots \text{(iii)}$

$\therefore \frac{\mathrm{m}_1\,\mathrm{m}_2}{\ell_1\ell_2}=\frac{\mathrm{c+a}}{\mathrm{b+c}}$

From equation (iii)

$\frac{\ell_1\ell_2}{b+c}=\frac{m_1m_2}{c+a}=\frac{n_1n_2}{a+b}=\frac{\ell_1\ell_2+m_1m_2+n_1n_2}{(b+c)+(b+c)+(a+b)}$

$\because$ lines are perpendicular $\quad\therefore\quad \ell_1\ell_2+m_1m_2+n_1n_2=0$

then $(b+c)+(c+a)+(a+b)$ must be zero $2a+2b+2c=0 \quad\Rightarrow a+b+c=0$

Q26 (2) - Paragraph 2

$\mathrm{u}\ell+\mathrm{vm}+\mathrm{wn}=0$

$\mathrm{a}\ell^2+\mathrm{bm}^2+\mathrm{cn}^2=0$

$\mathrm{a}\ell^2+\mathrm{bm}^2+\mathrm{c}\left\{-\frac{(4\ell+\mathrm{vm})}{\mathrm{w}}\right\}^2=0$

$\Rightarrow\left(aw^2+cu^2\right)\ell^2+\left(bw^2+cv^2\right)m^2+2\text{ cuv }\ell m=0$

$\Rightarrow\left(\mathrm{aw}^2+\mathrm{cu}^2\right)\left(\frac{\ell}{\mathrm{m}}\right)^2+\left(\mathrm{bw}^2+\mathrm{cv}^2\right)+2\,\mathrm{cuv}\left(\frac{\ell}{\mathrm{m}}\right)=0\ldots\ldots\ldots\ldots\text{(i)}$

put $u=v=w=1$ in equation, then

$(a+c)\left(\frac{\ell}{m}\right)^2+2c\left(\frac{\ell}{m}\right)+(b+c)=0$

similarly $(a+b)\left(\frac{m}{n}\right)^2+2a\left(\frac{m}{n}\right)+(c+a)=0$

and $(b+c)\left(\frac{n}{\ell}\right)^2+2b\left(\frac{n}{\ell}\right)+(a+b)=0\ldots\ldots\ldots\ldots\text{(ii)}$

From equation

(ii) $\frac{n_1}{\ell_1}\cdot\frac{n_2}{\ell_2}=\frac{a+b}{b+c}$

similarly $\frac{\ell_1\ell_2}{\mathrm{b+c}}=\frac{\mathrm{m}_1\,\mathrm{m}_2}{\mathrm{c+a}}=\frac{\mathrm{n}_1\mathrm{n}_2}{\mathrm{a+b}}\ldots\ldots\ldots\text{(iii)}$

$\therefore\quad\frac{\mathrm{m}_1\,\mathrm{m}_2}{\ell_1\ell_2}=\frac{\mathrm{c+a}}{\mathrm{b+c}}$

From equation (iii)

$\frac{\ell_1\ell_2}{b+c}=\frac{m_1m_2}{c+a}=\frac{n_1n_2}{a+b}=\frac{\ell_1\ell_2+m_1m_2+n_1n_2}{(b+c)+(b+c)+(a+b)}$

$\because$ lines are perpendicular $\quad\therefore\quad \ell_1\ell_2+m_1m_2+n_1n_2=0$

then $(b+c)+(c+a)+(a+b)$ must be zero $2a+2b+2c=0 \quad\Rightarrow a+b+c=0$

Q27 (2) - Paragraph 3

$\vec{a}_1=\left[(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}-6\hat{\mathrm{k}})\cdot\frac{(2\hat{\mathrm{i}}-3\hat{\mathrm{j}}+6\hat{\mathrm{k}})}{7}\right]\frac{2\hat{\mathrm{i}}-3\hat{\mathrm{j}}+6\hat{\mathrm{k}}}{7}=\frac{-41}{49}(2\hat{\mathrm{i}}-3\hat{\mathrm{j}}+6\hat{\mathrm{k}})$

$$\vec{a}_2=\frac{-41}{49}\left((2\hat{\mathrm{i}}-3\hat{\mathrm{j}}+6\hat{\mathrm{k}})\cdot\frac{(-2\hat{\mathbf{i}}+3\hat{\mathrm{j}}+6\hat{\mathrm{k}})}{7}\right)\frac{(-2\hat{\mathbf{i}}+3\hat{\mathrm{j}}+6\hat{\mathrm{k}})}{7}$$

$$=\frac{-41}{(49)^2}(-4-9+36)(-2\hat{\mathbf{i}}+3\hat{\mathrm{j}}+6\hat{\mathrm{k}})=\frac{943}{49^2}(2\hat{\mathbf{i}}-3\hat{\mathrm{j}}-6\hat{\mathrm{k}})$$

Q28 (1) - Paragraph 3

$\vec{a}_1 \cdot \vec{b} = \frac{-41}{49}(2\hat{i} - 3\hat{j} + 6\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) = -41$

Q29 (3) - Paragraph 4

$\overrightarrow{\mathrm{BL}} = \frac{1}{3}\overrightarrow{\mathrm{b}}$

$\therefore \quad \overrightarrow{\mathrm{AL}} = \overrightarrow{\mathrm{a}} + \frac{1}{3}\overrightarrow{\mathrm{b}}$

Let $\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{AL}}$ and P divides DB in the ratio $\mu : 1 - \mu$

Then $\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{a}} + \frac{\lambda}{3}\overrightarrow{\mathrm{b}}$ .... (i)

Also $\overrightarrow{\mathrm{AP}} = \mu\overrightarrow{\mathrm{a}} + (1-\mu)\overrightarrow{\mathrm{b}}$ .... (ii)

from (i) and (ii) $\lambda\vec{a} + \frac{\lambda}{3}\vec{b} = \mu\vec{a} + (1-\mu)\vec{b}$

$\therefore \quad \lambda = \mu$ and $\frac{\lambda}{3} = 1 - \mu$

$\therefore \quad \lambda = \frac{3}{4}$

$\therefore$ P divides A L in the ratio 3 : 1 and P divides DB in the ratio 3 : 1 similarly Q divides DB in the ratio 1 : 3

thus $\quad \mathrm{DQ} = \frac{1}{4}\mathrm{DB} \quad$ and $\quad \mathrm{PB} = \frac{1}{4}\mathrm{DB}$

$PQ = \frac{1}{2}DB$ i.e. $\quad PQ : DB = 1 : 2$

Q30 (2) - Paragraph 4

$\overrightarrow{\mathrm{BL}} = \frac{1}{3}\overrightarrow{\mathrm{b}}$

$\therefore \quad \overrightarrow{\mathrm{AL}} = \overrightarrow{\mathrm{a}} + \frac{1}{3}\overrightarrow{\mathrm{b}}$

Let $\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{AL}}$ and P divides DB in the ratio $\mu : 1 - \mu$

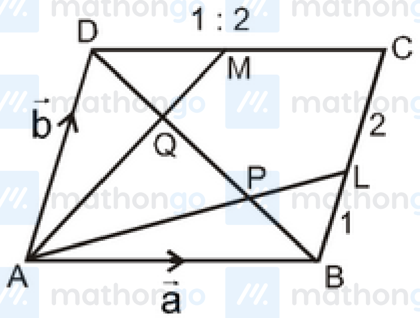
Then $\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{a}} + \frac{\lambda}{3}\overrightarrow{\mathrm{b}}$ .... (i)

Also $\overrightarrow{\mathrm{AP}} = \mu\overrightarrow{\mathrm{a}} + (1-\mu)\overrightarrow{\mathrm{b}}$ .... (ii)

from (i) and (ii) $\lambda\vec{a} + \frac{\lambda}{3}\vec{b} = \mu\vec{a} + (1-\mu)\vec{b}$

$\therefore \quad \lambda = \mu$ and $\frac{\lambda}{3} = 1 - \mu$

$\therefore \quad \lambda = \frac{3}{4}$



$\therefore$ P divides A L in the ratio 3 : 1 and P divides DB in the ratio 3 : 1 similarly Q divides DB in the ratio 1 : 3

thus $\mathrm{DQ} = \frac{1}{4}\mathrm{DB}$ and $\mathrm{PB} = \frac{1}{4}\mathrm{DB}$

$PQ = \frac{1}{2}DB$ i.e. $PQ : DB = 1 : 2$

Q31 (1) - Paragraph 5

The diagonals are $\overrightarrow{\mathrm{d}}_1 = 3\hat{a} - 2\hat{\mathrm{b}} + 2\hat{c} + (-\hat{\mathrm{a}} - 2\hat{c}) = 2\hat{\mathrm{a}} - 2\hat{\mathrm{b}}$

$\overrightarrow{\mathrm{d}}_2 = 3\hat{a} - 2\hat{\mathrm{b}} + 2\hat{\mathrm{c}} - (-\hat{\mathrm{a}} - 2\hat{\mathrm{c}}) = 4\hat{\mathrm{a}} - 2\hat{\mathrm{b}} + 4\hat{\mathrm{c}}$

Angle between them $= \cos^{-1}\frac{\overrightarrow{\mathrm{d}}_1 \cdot \overrightarrow{\mathrm{d}}_2}{\left|\overrightarrow{\mathrm{d}}_1\right| \cdot \left|\overrightarrow{\mathrm{d}}_2\right|} = \cos^{-1}\left(\frac{8+4}{2\sqrt{2}(6)}\right) = \cos^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$

Q32 (4) - Paragraph 5

$\overrightarrow{\mathrm{x}} + \overrightarrow{\mathrm{y}} = 2\hat{\mathrm{b}} - 3\hat{\mathrm{c}}$ and $\overrightarrow{\mathrm{y}} + \overrightarrow{\mathrm{z}} = -2\hat{\mathrm{a}} + 3\hat{\mathrm{b}} - 3\hat{\mathrm{c}}$

$$\therefore \quad (\vec{x} + \vec{y}) \times (\vec{y} + \vec{z}) = \begin{vmatrix} \hat{a} & \hat{b} & \hat{c} \\ 0 & 2 & -3 \\ -2 & 3 & -3 \end{vmatrix} = 3\hat{a} + 6\hat{b} + 4\hat{c}$$

$\therefore$ required unit vector $= \frac{3\hat{a}+6\hat{b}+4\hat{c}}{\sqrt{61}}$

Q33 (0) - Integer Type

Let equation of a plane containing the line be $\ell(x-1) + m(y+2) + nz = 0$ then $2\ell - 3m + 5n = 0$ and $\ell - m + n = 0$

$\therefore \frac{\ell}{2} = \frac{m}{3} = \frac{n}{1}$

the plane is $2(x-1) + 3(y+2) + z = 0$

i.e. $2x+3y+z+4=0$

$\therefore \quad a=2, b=-3, c=1$

Q34 (3) - Integer Type

$\vec{r}=3\vec{i}+8\vec{j}+3\vec{k}+\lambda(3\vec{i}-\vec{j}+\vec{k}) \quad \ldots\ldots\ldots\ldots$ (i)

$\vec{r}=-3\vec{i}-7\vec{j}+6\vec{k}+\mu(-3\vec{i}+2\vec{j}+4\vec{k}) \quad \ldots\ldots\ldots$ (ii)

Let L and M be points on the line (i) and (ii) respectively So that LM is perpendicular to both the lines. Let position vector of L be $3\vec{\mathrm{i}}+8\vec{\mathrm{j}}+3\vec{\mathrm{k}}+\lambda_0(3\vec{\mathrm{i}}-\vec{\mathrm{j}}+\vec{\mathrm{k}})$

and the position vector of M be $-3\vec{i}-7\vec{j}+6\vec{k}+\mu_0(-3\vec{i}+2\vec{j}+4\vec{k})$

then $\overrightarrow{\mathrm{LM}}=-6\vec{\mathrm{i}}-15\vec{\mathrm{j}}+3\vec{\mathrm{k}}-\lambda_0(3\vec{\mathrm{i}}-\vec{\mathrm{j}}+\vec{\mathrm{k}})+\mu_0(-3\vec{\mathrm{i}}+2\vec{\mathrm{j}}+4\vec{\mathrm{k}})$

since $\overrightarrow{\mathrm{LM}}$ is perpendicular to both the lines (i) and (ii)

$\therefore \quad \overrightarrow{\mathrm{LM}}\cdot(3\vec{\mathrm{i}}-\vec{\mathrm{j}}+\vec{\mathrm{k}})=0$ and $\overrightarrow{\mathrm{LM}}\cdot(-3\vec{\mathrm{i}}+2\vec{\mathrm{j}}+4\vec{\mathrm{k}})=0$

Thus $-18+15+3-\lambda_0(9+1+1)+\mu_0(-9-2+4)=0$

i.e. $-11\lambda_0-7\mu_0=0 \quad \ldots\ldots$ (iii)

and $18-30+12-\lambda_0(-9-2+4)+\mu_0(9+4+16)=0$

(iv) i.e. $7\lambda_0+29\mu_0=0 \quad \ldots\ldots$ (iv)

from (iii) and (iv) we get $\lambda_0=\mu_0=0$

$\therefore \quad \vec{\mathrm{M}}=-6\vec{\mathrm{i}}-15\vec{\mathrm{j}}+3\vec{\mathrm{k}}$

$\therefore \quad |\overrightarrow{\mathrm{LM}}|=\sqrt{36+225+9}=\sqrt{270}=3\sqrt{30}$

position vector of L is $3\vec{i}+8\vec{j}+3\vec{k}$

$\therefore$ equation of the line of shortest distance (LM) is $\vec{\mathrm{r}}=3\vec{\mathrm{i}}+8\vec{\mathrm{j}}+3\vec{\mathrm{k}}+\lambda(-6\vec{\mathrm{i}}-15\vec{\mathrm{j}}+3\vec{\mathrm{k}})$

Q35 (50) - Integer Type

$\mathrm{V}_1=[\vec{\mathrm{a}}\ \vec{\mathrm{b}}\ \vec{\mathrm{c}}]$

$\mathrm{V}_2=\frac{1}{2}[\vec{\mathrm{a}}\ \vec{\mathrm{b}}\ \vec{\mathrm{c}}]$

$\mathrm{V}_3=\frac{1}{6}[\vec{\mathrm{a}}\ \vec{\mathrm{b}}\ \vec{\mathrm{c}}]$

$\mathrm{V}_1:\mathrm{V}_2:\mathrm{V}_3=1:\frac{1}{2}:\frac{1}{6}$

$=6:3:1$

$$V_1=\begin{vmatrix}1 & -1 & -6\\ 1 & -1 & 4\\ 2 & -5 & 3\end{vmatrix}=1(-3+20)+1(3-8)-6(-5+2)=17-5+18=30$$

$\therefore V_1+V_2+V_3=30+15+5=50$

Q36 (2) - Integer Type

$\sum[\vec{p}\times\{(\vec{x}-\vec{q})\times\vec{p}\}]=\vec{0}$

$\Rightarrow \quad \sum[\vec{p}\times(\vec{x}\times\vec{p})]-\sum[\vec{p}\times(\vec{q}\times\vec{p})]=\vec{0}$

$\Rightarrow \quad \sum\vec{p}^{\,2}\vec{x}-\sum(\vec{p}\cdot\vec{x})\vec{p}-\sum\vec{p}^{\,2}\vec{q}+\sum(\vec{p}\cdot\vec{q})\vec{p}=\vec{0}$

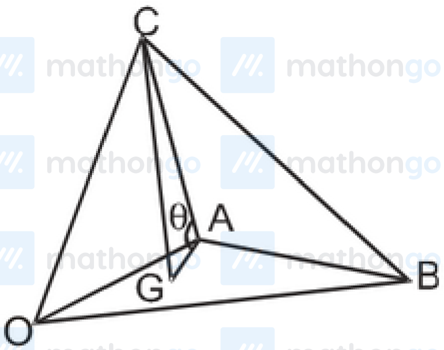
$\Rightarrow \quad 3\vec{p}^{\,2}\vec{x}-\vec{p}^{\,2}\vec{x}-\vec{p}^{\,2}(\vec{p}+\vec{q}+\vec{r})=\vec{0}$

$\Rightarrow \quad 2\vec{p}^{\,2}\vec{x}=\vec{p}^{\,2}(\vec{p}+\vec{q}+\vec{r})$

$\Rightarrow \quad \vec{x}=\frac{1}{2}(\vec{p}+\vec{q}+\vec{r})$

Q37 (13) - Integer Type

Let OABC be the tetrahedron. Let G be the centroid of the face OAB, then $GA=\frac{1}{\sqrt{3}}AC$.



Then $\cos\theta=\frac{GA}{CA}=\frac{1}{\sqrt{3}}$

$\therefore \quad \cos^2\theta=\frac{1}{3}$

$\therefore \quad a=1$ and $b=3$

$\therefore \quad 10a+b=13$